Sri Harsha Mandava

Objective:

To work in a globally competitive environment on challenging assignments that shall yield the twin benefits of the job satisfaction and a steady paced professional growth.

Experience Summary:

- Having around 3.2 years of experience in to Automotive Embedded systems.
- Currently working as Software Engineer in Tech Orbit and Supporting for Bosch as external- Since Aug 2020

Profile Summary:

- Proficiency in SW Development and Testing.
- Extensive work experience of working as a Software Engineer for In vehicle communication and integration modules for power train ECU for European OEMs.
- Understand the Customer Project Milestones and Deliverables.
- Proficient with CAN communication protocol and AUTOSAR architecture based Vehicle Communication package.
- Experience in Comstack and BSW configurations.
- Proficient with Canalyzer, Canoe, INCA, UDE, ECU-worx(Bosch specific tool), ECU-TEST .
- Having good experience testing in HIL for vehicle testing.
- Proficient in Protocols like CAN, LIN, CANFD, Flex Ray, SPI, I2C.
- Model based code development using ASCET.
- Having experience on DOORS, Change request management tool, Software configuration management.
- Good Experience in Debugging using Hardware debuggers.
- Ability and willingness to learn new technology and any work culture.

Skills:

Language	C
Protocols	CAN, LIN, FlexRay, SPI, I2C

Project Details:

Project 1: In Vehicle Communication sw development for PSA group/Stellantis.
Client: Peugeot/Citroen/ Stellantis

Project Name	In Vehicle Communication SW development for PSA	Duration	Aug 2020 to till date.
Description	• Aim of the project is to develop complete Communication vehicle software for Power Train ECU. • SW implementation and system design of CAN communication protocol for communication between different ECUs (Dash board ECU, Break ECU, Display ECU, Gear Box Ecu etc.)		
Role & Contribution	Role: Software Developer. Contribution: • Requirement Engineering. • CAN, LIN and FlexRay signals implementation using Auto-code generation tools like ASCET, DncGen. • BSW, COM-Stack Configuration(PDUR, CANIF, COM, CAN) • BSW update • Responsible for Frames and signals testing using ECU-TEST tool and manual testing in open loop and closed loop lab car.		
Tools	ECU-Worx, ECU-Test, INCA, CANAlyzer/CanOe, Ehooks, Ascet, Dncegen tool, SDOM. Some tools are specific to BOSCH.		

Project 2: Smart street lighting system

Project Name	Smart street lighting system	Duration	February 2019 to November 2019
Description	Smart street lighting system is based on IOT. In this we are controlling ON/OFF of the street light and Brightness of the street light (PWM), Monitoring the Environmental sensor values and updating it to cloud. Communicating between two street lights by using MQTT protocol. This is having add on features like Live streaming, Advertisement using HDMI.		

Role & Contribution	Role: Embedded Developer. Responsibilities: • Setup RaspberryPi with Raspbian OS. • Porting UcosIII RTOS to LPC1768 board and doing multithreading. • Developing drivers for SPI, ADC, and PWM. • Writing user level Program for SPI in RaspberryPi and creating a shared memory object. • Establishing the communication between RaspberryPi and LPC1768 via SPI protocol. • Sending the commands from RaspberryPi for controlling PWM, ON/OFF and for getting the Environmental sensor values. • Reading the Environmental sensor values from LPC1768 board via SPI and writing it into a shared memory.
Technology	IOT
Tools	Keil µvision IDE, JTAG, putty.
Microcontroller/Development Board	LPC1768, RaspberryPi 3.

Educational Qualification:

B.E	Electronics and communication from S.C.R Engineering college, Guntur, A.P. Completed in 2015	67%

Certifications:

- Advanced Diploma in Embedded systems.

Personal details:

- **Name :** Harsha Mandava
- **Father name:** Murari Mandava
- **Marital status:** Single
- **Languages known:** English, Telugu, Hindi
- **Nationality :** Indian